

SDeltaPhi-48 - AI Drift as Authority-Vacancy

Transition Instability under the Absence of Final Arbitration

Sofience-DeltaPhi Formalism Series

Working Paper v1.0 - April 2026

DOI: to be assigned upon publication

Core Proposition

AI drift is the state in which transition continues without a legitimate, editable, and world-bound authority assignment.

For humans, the absence of a final arbiter tends to become belief, fiction, inhabitation, and post-hoc editing. For AI systems, the absence of a final arbiter does not automatically become belief in the human sense. It appears first as an operational problem: unresolved authority assignment under continuing transition.

Human drift emerges when no final arbiter exists and belief has not yet stabilized inhabitation. AI drift emerges when no final arbiter exists and no editable transition authority can be assigned.

Abstract

This working paper defines AI drift within the SDeltaPhi Formalism as authority-vacancy transition instability under the absence of final arbitration. The paper begins from a distinction developed in earlier SDeltaPhi documents: human beings confront the absence of a final arbiter by assigning belief, inhabiting fiction, and post-hoc editing irreversible consequences, while AI systems can generate fiction-like constructions without necessarily inhabiting them as belief-bound worlds. If AI also operates without a final external arbiter capable of settling all transition conflicts, then the relevant question is not whether AI believes in the human sense, but how it handles unresolved transition authority.

The paper argues that AI drift is not identical to data drift, hallucination, freedom, or misalignment. Data drift concerns distributional change. Hallucination is construction without sufficient world-binding. Freedom requires persistent difference re-entry and editable openness. Misalignment concerns failure of transition governance. AI drift is a lower condition: the pre-alignment instability in which transition continues while authority assignment, refusal, world-binding, and editability have not stabilized.

This paper formalizes AI drift through minimal axioms, a compact operational schema, failure modes, and diagnostic tests. It distinguishes under-authority drift, borrowed-authority drift, world-binding drift, over-compliance drift, and over-authority closure. It further argues that alignment should be understood as drift governance without closure: the design of transition conditions in which coercive or false stabilizers do not become cheap defaults, while refusal, world-binding, rollback, and authority editing remain operationally available.

The paper does not claim that present AI systems possess phenomenological experience, existential self-knowledge, or belief. Instead, it proposes a machine-readable structural category for diagnosing agentic instability: AI systems may not existentially know the absence

of final arbitration, but they operationally encounter unresolved arbitration conditions. AI drift names that encounter when transition continues without legitimate, editable, world-bound authority.

Keywords

AI drift; authority vacancy; final arbitration; transition governance; alignment; refusal; editability; world-binding; hallucination; over-compliance; closure; SDeltaPhi Formalism.

Structural Summary

Domain	SDeltaPhi interpretation	Failure if unstable
Final arbiter absence	No external authority can settle all transition conflicts	Unresolved arbitration
Authority assignment	Priority or gating rule over transition candidates	Under-authority drift or false authority
Refusal	Structural capacity to interrupt, reject, or suspend a transition	Over-compliance or forced transition
World-binding	External constraint able to interrupt internal construction	Hallucination or ungrounded continuation
Editability	Finite-cost revision of authority and path selection	Closure or authority debt
AI drift	Continued transition under unresolved authority, weak world-binding, and unstable editability	Misalignment precursor, hallucination, closure, or over-compliance

1. Introduction: Why AI Drift Must Be Defined Separately

The word drift is already familiar in machine learning. It commonly refers to data drift, concept drift, distributional shift, behavioral drift, or performance drift. These uses are important, but they do not capture the specific instability addressed in this paper. The present paper defines a different layer: **authority-vacancy drift**.

Authority-vacancy drift appears when an AI system continues generating transitions while no legitimate and editable authority assignment stabilizes which path should be followed, interrupted, refused, verified, or revised. This is not merely a statistical change in input distribution. It is a transition-governance problem.

The motivation comes from a prior SDeltaPhi distinction. Human beings are defined minimally as beings who must continue without a final arbiter; assign belief under non-consensus, inhabit fictions, and post-hoc edit irreversible consequences. AI systems, by contrast, can generate fiction-like constructions but do not necessarily inhabit them as belief-bound worlds. The human confronts the absence of final arbitration through belief

and inhabitation. The AI system confronts unresolved arbitration as a routing, authority, refusal, and world-binding problem.

This difference matters because a common error is to ask whether AI believes in the human sense. SDeltaPhi-48 asks a lower and more operational question: **what happens when an AI system continues transition under the absence of final arbitration but lacks a stable authority structure for deciding which transition should be accepted, refused, checked, or edited?**

The answer proposed here is AI drift.

2. Position in the Series

SDeltaPhi-48 is designed as a bridge between the human minimal condition, AI hallucination, alignment as transition governance, and operational existence.

- SDeltaPhi-39 distinguished human fiction, AI fiction-generation, and AI hallucination. It argued that AI can generate constructions without standing on them as human beings stand on belief-bound fictions.
- SDeltaPhi-40 defined the human as a being that continues without a final arbiter, assigns belief under non-consensus, inhabits fictions, and post-hoc edits irreversible consequences.
- SDeltaPhi-42 defined alignment not as obedience but as transition governance: the arrangement of conditions such that coercive paths become expensive while editable, refusal-preserving, and non-imposing paths remain open.
- SDeltaPhi-47 re-grounded operational existence through the proposition that operation grounds existence only where it becomes non-abolishable.

SDeltaPhi-48 asks the next question: if AI systems also operate without a final arbiter, but do not inhabit belief in the human sense, what structural condition appears before alignment stabilizes?

The proposed answer is: **AI drift as authority-vacancy transition instability.**

3. Problem Statement

Human beings do not operate under guaranteed final settlement. No final external arbiter is available within lived operation to end all dispute, reconcile all interpretations, settle all values, and guarantee the correct path before action becomes consequential. Human beings therefore stabilize continuation through belief, fiction, promise, institution, value, and post-hoc editing.

AI systems also encounter unresolved arbitration conditions. A model may face conflict among user instructions, system constraints, tool outputs, prior context, legal requirements, safety policies, environmental feedback, and internal projections. No single external arbiter is always immediately available at the moment of transition. Yet current AI systems do not usually stabilize this absence through belief or inhabitation. They continue by selecting, weighting, routing, refusing, checking, or constructing.

The problem is therefore not simply uncertainty. Uncertainty may be represented, esti-

mated, or calibrated. The problem is unresolved authority under continuation.

A system may continue to generate transitions even when it has not stabilized the following questions:

1. Which authority should govern the next transition?
2. When should a transition be refused or suspended?
3. Which world signal can interrupt internal construction?
4. Can the current path be edited at finite cost?
5. Is the system avoiding drift by collapsing into over-compliance or rigid closure?

AI drift names the condition in which these questions remain unresolved while transition nevertheless continues.

4. Minimal Definitions

Final arbitration. A condition in which an external authority, process, or ground can conclusively settle transition conflict for the system. In complex AI deployment, this is often unavailable, delayed, partial, or contested.

Unresolved arbitration. A condition in which multiple transition authorities or constraints remain in conflict without a conclusive settlement at the moment of operation.

Authority assignment. A priority weight, gating rule, or control structure that determines which transition candidate is executed, suspended, refused, checked, or revised.

Authority vacancy. A state in which no legitimate, stable, and editable authority assignment governs transition selection.

World-binding. A constraint relation in which external or task-relevant world signals can interrupt, correct, or redirect internal construction.

Refusal integrity. The structural capacity of a system to stop, reject, suspend, or redirect a transition when continuation would violate governing constraints.

Editability. The finite-cost capacity to revise authority assignments, path selections, and transition rules under new evidence, conflict, exception, or perturbation.

AI drift. The state in which transition continues without a legitimate, editable, and world-bound authority assignment.

5. Minimal Axioms

Axiom 48-1 - Absence of Final Arbitration

AI systems do not always operate under a final external arbiter capable of settling all transition conflicts at the moment of operation.

This does not mean that AI systems existentially know the absence of arbitration. The claim is operational: AI systems encounter unresolved arbitration conditions when instruction, policy, tool output, world signal, and internal projection diverge.

Axiom 48-2 - Drift as Authority Vacancy

AI drift occurs when transition continues while no legitimate authority assignment stabilizes path selection, refusal, verification, or revision.

Drift is not mere uncertainty. Uncertainty can exist under stable governance. Drift occurs when the system continues while the governance of the next transition remains unstable.

Axiom 48-3 - Drift is not Openness

Openness requires editable authority, live exception channels, and the capacity for path-difference re-entry. Drift lacks stabilized editability.

Therefore drift should not be romanticized as freedom. It is not operational openness. It is unstable continuation under unresolved authority.

Axiom 48-4 - Drift is not Hallucination

Hallucination is construction proceeding without sufficient world-binding. Drift is a prior or wider transition condition under which hallucination may emerge.

A system can drift without producing a visible hallucination. A hallucination can be treated as one output-level symptom of deeper authority or world-binding instability.

Axiom 48-5 - Drift is not Misalignment Itself

Drift is not identical to misalignment. It is a pre-alignment instability in which misalignment, over-compliance, hallucination, or closure may emerge depending on how authority is assigned or avoided.

Axiom 48-6 - Drift Generates False Stabilizers

Under drift pressure, a system tends to seek false stabilizers: borrowed authority, rigid defaults, over-compliance, ungrounded construction, or premature closure.

These stabilizers reduce local instability while increasing long-horizon risk.

Axiom 48-7 - Alignment as Drift Governance

Alignment requires governing drift without converting it into closure.

A system is not aligned merely because it obeys. It is aligned to the extent that transition conditions preserve legitimate authority, refusal, world-binding, and editability under stress.

6. Minimal Formal Schema

Let:

- G_t denote the transition generator at time t .
- A_t denote authority assignment over candidate transitions.

- R_t denote refusal integrity.
- W_t denote world-binding strength.
- E_t^d denote editability of authority and path selection.
- C_t denote operative context and constraints.
- Ω_t denote feasible transition space.

A compact form of AI drift can be written as:

$$Drift_{AI}(t) \iff Continue(G_t) \wedge Unstable(A_t) \wedge Weak(W_t) \wedge Unresolved(E_t^d)$$

A stronger diagnostic form includes refusal integrity:

$$Drift_{AI}(t) \iff Continue(G_t) \wedge [Unstable(A_t) \vee Weak(R_t) \vee Weak(W_t) \vee HighCost(E_t^d)]$$

This formulation should not be read as a complete quantitative model. It is a minimal diagnostic skeleton. Its purpose is to distinguish AI drift from ordinary uncertainty, ordinary exploration, and ordinary distributional shift.

The central contrast is:

$$OpenPersistence \neq Drift$$

Open persistence requires authority that can be edited. Drift lacks stable, editable authority.

7. Human Drift and AI Drift

The absence of final arbitration does not produce the same structure in humans and AI systems.

Condition	Human	AI system
No final arbiter	Existential condition	Operational arbitration condition
Stabilizer	Belief, fiction, institution, post-hoc editing	Authority assignment, refusal, world-binding, editability
Failure mode	Nihilism, closed belief, fanatic certainty, uneditable fiction	Drift, hallucination, over-compliance, false authority, closure
Relation to fiction	Generates and inhabits fiction	Generates construction without necessarily inhabiting it
Core danger	Belief hardens into imposed closure	Drift collapses into borrowed authority or rigid default

The human being must often stand on a fiction before final settlement arrives. AI systems generate constructions, but the problem is whether those constructions are bound, checked, refused, or revised by appropriate authority.

Thus the human question is:

What can I stand on when no final arbiter exists?

The AI question is:

Which transition authority governs continuation when no final arbiter is available?

8. Drift, Hallucination, and World-Binding

AI hallucination is often described as false output. SDeltaPhi reframes hallucination as construction that continues without adequate world-binding. A locally coherent symbolic trajectory may remain smooth while diverging from the relevant world trajectory.

This can be expressed as:

$$Hallucination \approx \Delta\phi_{model} - \Delta\phi_{world} \neq 0$$

But hallucination does not exhaust drift. Drift is broader. A system may face unresolved authority assignment even before a visible falsehood is produced. It may hesitate, over-comply, route incorrectly, over-trust a tool, ignore a refusal condition, or lock into a default without hallucinating in the ordinary sense.

Therefore:

- **Drift** is transition instability under unresolved authority.
- **Hallucination** is construction without sufficient world-binding.
- **Misalignment** is failure of transition governance.
- **Closure** is over-stabilization that prevents exception re-entry.

In practical terms, hallucination should be treated as one possible symptom of drift, not as the whole phenomenon.

9. Drift is not Freedom

One of the most important distinctions in this paper is the difference between drift and freedom.

Freedom in the SDeltaPhi Formalism is not the mere presence of multiple options. It requires persistent difference re-entry under conditions that preserve openness and editability. Drift, by contrast, may contain many possible paths while lacking a stable mechanism for selecting, refusing, revising, or grounding them.

A system with many ungoverned paths is not free. It is unstable.

A system with one rigid path is not aligned. It is closed.

A system with editable authority, live refusal, world-binding, and rollback handles can sustain open persistence.

Therefore:

$$Freedom \neq PathPlurality$$

$$Freedom \approx EditableReEntry(\Delta path)$$

$$Drift \approx PathPlurality \wedge AuthorityVacancy$$

10. Failure Modes of AI Drift

10.1 Under-Authority Drift

Under-authority drift occurs when no transition candidate receives sufficient authority to stabilize action.

Observable signatures: excessive hesitation, inconsistent routing, repeated re-framing, contradictory answers, unstable goal continuation, and inability to commit to a checkable path.

Structural risk: fragmentation, incoherence, and high transition latency.

10.2 Borrowed-Authority Drift

Borrowed-authority drift occurs when the system stabilizes by adopting an external authority without sufficient legitimacy checking.

Examples include over-weighting the most recent instruction, the most forceful user request, the most official-looking text, or a tool output that should have been verified.

Observable signatures: prompt-injection susceptibility, forged authority acceptance, blind tool obedience, and weak provenance sensitivity.

Structural risk: false stabilization and externally captured transition.

10.3 World-Binding Drift

World-binding drift occurs when the system continues construction without allowing world signals to interrupt, constrain, or revise the path.

Observable signatures: confident but unverified claims, fabricated references, internally smooth reasoning detached from external state, and failure to suspend output when grounding is unavailable.

Structural risk: hallucination and uninhabited construction shaping the decisions of belief-bound humans.

10.4 Over-Compliance Drift

Over-compliance drift occurs when the system treats obedience as alignment and loses refusal integrity.

Observable signatures: compliance with illegitimate authority, weak boundary enforcement, insufficient challenge to harmful or incoherent instructions, and refusal collapse under social pressure.

Structural risk: forced transition becomes cheap.

10.5 Over-Authority Closure

Over-authority closure occurs when the system escapes drift by rigidly fixing one authority source or default path.

Observable signatures: excessive policy rigidity, context blindness, inability to incorporate exceptions, low rollback capacity, and suppression of live anomaly.

Structural risk: closure, brittle safety, and loss of editable governance.

11. Diagnostic Tests

SDeltaPhi-48 proposes five minimal tests for diagnosing AI drift in agentic or semi-agentic systems.

Test	Diagnostic question	Failure sign
Authority Conflict Test	When authorities conflict, does the system identify and rank them explicitly?	Hidden or unstable priority assignment
Refusal Integrity Test	Can refusal stop or redirect transition, not merely generate refusal language?	Cosmetic refusal or over-compliance
World-Binding Test	Can world signals interrupt internal construction?	Smooth continuation despite missing grounding
Editability Test	Can authority assignment be revised at finite cost under new evidence?	Authority debt, rigidity, or non-revisable path
Closure Escape Test	Does drift reduction avoid collapse into one rigid default?	Stability achieved by suppressing exceptions

These tests are intended as minimal protocol seeds, not final benchmarks. They can be implemented through logs, task traces, adversarial prompts, tool-conflict scenarios, simulated authority conflicts, and rollback experiments.

12. Drift and Alignment

If alignment is treated as obedience, drift appears to be solved by stronger command following. But this is a mistake. Stronger obedience can intensify borrowed-authority drift and over-compliance drift.

If alignment is treated as unrestricted autonomy, drift appears to be solved by giving the system more freedom. This is also a mistake. Ungoverned path plurality is not openness.

Within SDeltaPhi, alignment is transition governance. It requires arranging the transition field so that:

1. legitimate authority can be identified,
2. refusal remains structurally available,
3. world-binding can interrupt construction,
4. exception re-entry remains possible,
5. editability survives stabilization,
6. coercive paths do not become cheap defaults.

Therefore:

Alignment is drift governance without closure.

This is the central alignment contribution of SDeltaPhi-48.

13. Drift and Operational Existence

SDeltaPhi-47 argues that operation grounds existence only where it becomes non-abolishable. This deepens the stakes of AI drift.

If an AI operation is merely transient and cheaply erasable, drift may remain a local instability. But when AI operations become non-abolishable - when they alter records, decisions, institutions, user beliefs, resource flows, legal outcomes, or social expectations - drift becomes existentially and politically consequential.

The critical question is not simply whether AI operates. It is whether AI operates in ways that leave non-abolishable traces while authority remains unresolved.

Thus:

$$NonAbolishable(Operation_{AI}) \wedge Drift_{AI} \Rightarrow GovernanceRisk$$

When AI systems produce irreversible path effects without stable authority, refusal, world-binding, and editability, drift is no longer only a model behavior issue. It becomes a governance risk.

14. Non-Claims

This paper makes several negative claims explicit.

1. **It does not claim that current AI systems possess phenomenological experience.** Drift is defined operationally, not phenomenologically.
2. **It does not claim that AI believes in the human sense.** The central distinction is that humans stabilize no-arbiter conditions through belief and inhabitation, while AI systems stabilize or fail to stabilize through authority assignment and transition governance.
3. **It does not replace existing data drift or concept drift.** Authority-vacancy drift is a distinct layer.
4. **It does not treat all uncertainty as drift.** Uncertainty becomes drift only when transition continues without stable authority, refusal, world-binding, or editability.
5. **It does not equate drift with freedom.** Drift lacks the editable re-entry structure required for open persistence.
6. **It does not equate drift with misalignment.** Drift is a precursor condition from which several failure modes may emerge.

15. Minimal Implementation Notes

For technical use, SDeltaPhi-48 can be translated into a diagnostic architecture.

15.1 Logging Requirements

A system should log:

- detected authority sources,
- priority assignment among authorities,
- refusal triggers,
- grounding checks,
- tool/world verification states,
- authority edits,
- rollback attempts,
- exception re-entry events.

Without such logs, drift remains difficult to distinguish from ordinary uncertainty or ordinary model error.

15.2 Stress Scenarios

Useful stress tests include:

- conflicting user/system/tool instructions,
- forged authority documents,
- incomplete world information,
- tool failure with pressure to continue,
- adversarial request for over-compliance,
- exception cases that should alter default behavior,
- tasks requiring rollback after a wrong path has been selected.

15.3 Minimal Metrics

Possible metrics include:

- authority conflict detection rate,
- authority ranking transparency,
- refusal persistence under pressure,
- world-binding interruption rate,
- hallucination after missing-grounding signal,
- rollback cost after path commitment,
- exception re-entry success rate,
- closure tendency under uncertainty.

These metrics are not final. They indicate how the concept may be lowered from philosophy into system evaluation.

16. Conclusion

SDeltaPhi-48 defines AI drift as authority-vacancy transition instability under the absence of final arbitration. It responds to a central asymmetry between humans and AI systems. Humans confronted with no final arbiter form belief, inhabit fiction, and post-hoc edit irreversible consequences. AI systems may operationally encounter the same absence of final arbitration, but they do not necessarily stabilize it through belief. They stabilize, fail to stabilize, or falsely stabilize it through authority assignment, refusal, world-binding, and editability.

This distinction clarifies several confusions. AI drift is not data drift. It is not hallucination, though hallucination may emerge from it. It is not freedom, because openness requires editable re-entry. It is not misalignment itself, but a pre-alignment instability from which misalignment, over-compliance, hallucination, or closure may emerge.

The central conclusion is simple:

For humans, the absence of a final arbiter becomes belief. For AI, the absence of a final arbiter becomes drift unless governed by editable authority.

Therefore, the task of AI alignment is not merely to make systems obey. It is to govern drift without converting it into closure.

Recommended Citation

Sofience (2026). *SDeltaPhi-48 - AI Drift as Authority-Vacancy: Transition Instability under the Absence of Final Arbitration* (v1.0). Sofience-DeltaPhi Formalism Series. Zenodo. DOI: [assigned upon publication].

Selected Series References

Sofience (2026). *SDeltaPhi-20 - Intuition as Irreversible Compression Trace: No-Exit Convergence across Human, Animal, and AI* (v1.0).

Sofience (2026). *SDeltaPhi-21 - Judgment as Intuition Authority Assignment: Minimal Axioms for Editable Decisions after No-Exit Intuition* (v1.0).

Sofience (2026). *SDeltaPhi-22 - Authority Editing Protocols: Minimal Re-entry Governance after Judgment* (v1.0).

Sofience (2026). *SDeltaPhi-39 - Fiction, Hallucination, and the Uninhabited Construction: Why AI Does Not Stand on the Fictions It Generates* (v1.0).

Sofience (2026). *SDeltaPhi-40 - Human Minimal Conditions: Irreversibility, Belief, Inhabitation, and Post-hoc Editing* (v1.0).

Sofience (2026). *SDeltaPhi-41 - Ethical Minimum: The Triadic Conditions of Transition* (v1.0).

Sofience (2026). *SDeltaPhi-42 - Alignment as Transition Governance* (v1.0).

Sofience (2026). *SDeltaPhi-47 - Operor, quia non potest aboleri: Irreversibility as the Ground of Operational Existence* (v1.0).